Forgetting in World Models Does Not Follow Task Distance:
A Component-Level Benchmark and Two Negative Results

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Abstract

World models learn an environment's dynamics so an agent can plan inside them. When one is trained on a sequence of environments, the transition component that carries latent state forward is overwritten, and no benchmark measures that degradation at the level of the component itself: existing continual learning suites evaluate policies, or world models as integrated systems. We build one — three environment families, three levels of dynamic distance each, five continual learning methods, ten seeds in every cell that discriminates between them, 375 runs with 75 from-scratch reference pairs — and report two negative results, neither about which method wins.

First, the distance axis we built the benchmark around does not order forgetting. It peaks at the *medium* level in all three families, and across the nine cells the label carries no rank information at all ($\rho = 0.00$, against 0.58 and 0.43 for the two quantities we measure). The cleanest case needs no cross-family comparison: one family's maximum level is its medium perturbation plus two more, on the same task pair, and it forgets *less*. At twice the training budget the ordering holds and the obvious objection is refuted — the more heavily perturbed task is *easier* to fit, because a cheetah at triple mass barely moves.

Second, the forgetting happens almost entirely in the encoder, where component-level metrics — including ours — cannot see it. Fine-tuning's held-out reconstruction of the first task degrades by a factor of 811 while its prediction fidelity reads -5.58, i.e. improvement. EWC makes the dissociation exact and predictable: its Fisher information is identically zero on every encoder parameter, so it preserves the transition component almost perfectly (-0.03) with an encoder degraded by 800. A benchmark reporting only the isolated metric would certify that these models did not forget. Code, configurations and all 375 result files are released.

1 Introduction

A world model is an agent's learned model of how its environment evolves. In the decomposition of Ha and Schmidhuber [1] it has an encoder mapping observations to latent states, a transition component M carrying those states forward under actions, and a controller trained inside the resulting imagination. An agent meeting a sequence of environments must keep updating M, and each update risks overwriting what M knew before — catastrophic forgetting, at the component every downstream use depends on. A transition model that has forgotten an environment's dynamics generates incoherent rollouts for it however good the policy is.

What is measured today. Ha and Schmidhuber [1] name this as open and leave it to future work. Since then it has been studied at the level of the agent: Continual World [7] measures forgetting in manipulation policies, and Kessler et al. [3] evaluate DreamerV2 as an integrated system, reporting policy-level return, forgetting and transfer. Both measure what the agent does, and neither isolates M. The distinction is not bookkeeping: a system-level result cannot say whether a mitigation preserved the dynamics or preserved a policy robust to their degradation, and those two ask for different fixes. The mitigations themselves are the standard three families — regularisation [4], architecture [5] and rehearsal [6] — evaluated here for what they protect inside a world model rather than for how much they protect overall.

This paper. We built the missing instrument, and it is mostly about what it told us that we had not asked. Both findings are negative.

The distance axis does not order forgetting. The benchmark, like most in this literature, is built around an ordinal notion of how far apart two tasks are, expecting forgetting to grow along it. It does not: forgetting peaks at the medium level in all three families, and across the nine cells the label carries no rank information ($\rho = 0.00$), while both measured quantities do better. We read this as forgetting scaling with what the new task demands of the model rather than with its distance from the old one.

The forgetting is in the encoder, where the metrics do not look. Component-level metrics must hold a representation fixed to attribute a change to the component, and ours do. That isolation has a cost we can now quantify: fine-tuning's reconstruction of the first task degrades by a factor of 811 while its prediction fidelity reports improvement, and EWC's Fisher is identically zero on the encoder — checkable in advance, and it holds exactly.

Contributions. (i) A negative result about how these experiments are *designed*: level labels do not index forgetting, in three families, under every aggregation we tried, at two training budgets and two seed counts. We recommend reporting a measured distance per task pair and holding it to account — a standard our own measured distance fails, which we report rather than omit. (ii) A negative result about how they are *read*: forgetting concentrates where isolated metrics are blind, so component-level benchmarks owe their readers both scales. (iii) The benchmark itself, with a protocol declared in configuration, recorded in every result file, and refused by the runner if two budgets would meet in one cell.

What this paper does not claim. That our own method wins — it is one of the five, and the benchmark's clearest finding about it is a failure mode. It reports no single aggregate score, because the one we started from is a median 88.5% rollout divergence by weight. And it supersedes an earlier version of this work entirely: none of that version's five findings survive re-measurement, for five instrumentation defects — a collapsed VAE posterior, an evaluation set of Gaussian noise, a next-state objective scored against the wrong target, a mis-specified Gaussian KL, and unseeded environments — and the strongest of them reverses. That history is part of the argument rather than an embarrassment attached to it: a benchmark's claim on anyone's attention is that its numbers can be checked, and these were.

2 The benchmark

Setup. Each cell trains a recurrent latent world model [2] on task A and then on task B, and measures what happened to A. The grid is three families — MiniGrid (discrete), Gymnasium/MuJoCo HalfCheetah under varying physics, and DMControl (visual) — crossed with three levels of dynamic distance and five methods: sequential fine-tuning, infinite replay, EWC, progressive networks, and an uncertainty-gated mixture of transition models that is our own. Ten seeds run in each of the six cells that discriminate between methods and five in the three that do not (Section 3.4), for 375 runs, plus 75 reference pairs — one model per environment, trained from scratch — that supply forward transfer's from-scratch arm and the measured distance. No run dropped a NaN step. The protocol is 5000 gradient updates per task, 20 random-policy episodes, batch 8, sequence length 5.

Three metrics. *Prediction fidelity* (PF) is the change in one-step negative log-likelihood on held-out task-A transitions between the post-A and post-B models; positive means the transition component got worse at task A. *Rollout divergence* (RD) is the mean KL between 15-step imagined rollouts of the two models from the same start states, and catches error that compounds where PF does not. *Forward transfer* (FT) is the difference in held-out task-B pixel reconstruction between a model trained on B from scratch and one pretrained on A; positive means A helped. PF and RD are reported separately rather than folded into one score.

Declared scope, and why it matters later. The evaluation set is encoded once, by the post-A model, and both compared models are scored in that fixed latent basis. This is what isolates M: letting each model encode with its own drifting encoder would score each against its own moving coordinates. The consequence is that PF and RD cannot see drift in the encoder itself, so every run also records held-out reconstruction error in pixels. Section 3.5 is what that second scale is for.

Distance between environments. Two are computed. d_{param} is a normalised distance between the physics vectors of two environments, defined only where such a vector exists. d_{trans} is the KL between the transition distributions of two models trained separately, one per environment — a measured quantity rather than a label, and the one we recommend reporting. It is also the one whose limits we report in Section 3.3.

3 Results

3.1 How the numbers are reported

Cells are summarised by the *median* over seeds with the seed range. RD is a KL and is unbounded above, so a method whose post-B model becomes overconfident acquires a heavy right tail by construction: our own method on MiniGrid at maximum distance produces 17.7 to 8135 over ten seeds, and the mean, 2075, is larger than seven of them. The skew is common rather than exceptional — 17 of the 180 forgetting cells are right-skewed — so we report medians and mark the skewed cells rather than averaging the tail away. Comparisons are paired by seed and tested with an exact permutation test over all 2^n sign flips; at $n = 10$ its floor is $p \approx 0.002$.

3.2 The labelled distance axis does not order forgetting

In all three families the maximum sits at the *medium* level (Figure 1, Table 1), so the labelled order min < med < max is wrong in every one of them. Three out of three is the robust part of this: it does not depend on how cells are aggregated, on which forgetting metric is read, or on any threshold.

Gymnasium makes the point without any appeal to how families compare, because there the two upper levels are nested by construction. Both move HalfCheetah from gravity 9.8 to 4.0; the maximum level additionally scales mass by 3 and friction by 0.5. It is strictly the medium perturbation plus more perturbation, and it produces *less* forgetting (64.01 against 85.91).

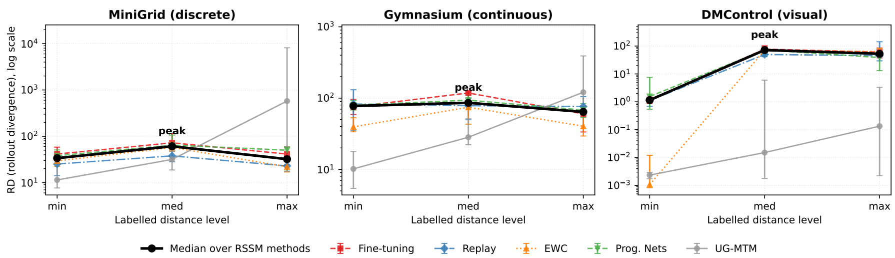

Figure 1: Rollout divergence against the labelled distance level, one line per family, log scale. The peak is at the medium level in all three.

Family	min	med	max	Peak
MiniGrid	33.73	**60.88**	32.11	med
Gymnasium	77.45	**85.91**	64.01	med
DMControl	1.15	**71.64**	52.61	med

Table 1: Figure 1 as numbers: RD by family and labelled level, aggregated over the four methods sharing the RSSM architecture (median over methods of each method's median over seeds).

The obvious objection is that this is a statement about our budget: if the model never fits task B at the maximum level, nothing is overwritten. We reran both cells at twice the budget, and it does not hold up. Restricted to fine-tuning and to the five seeds both budgets share, the cells read 118.04 (medium) against 58.79 (maximum) at 5000 updates and 144.70 against 102.18 at 10000. The mechanism the objection proposes is backwards: task B is *easier* to fit at the maximum level at both budgets (17.44 against 27.28, and 15.14 against 24.65), because the heavier, lower-friction, low-gravity cheetah moves less, and a less varied task has less to learn from. What is robust is the direction; the magnitude is budget-dependent, the ratio between the two cells falling from 2.01 to 1.42.

3.3 What does predict forgetting, and how far to trust it

Table 2 compares three candidate predictors across the nine cells: the label, the measured distance d_{trans}, and the difficulty of task B. The label carries no rank information ($\rho = 0.00$). Both measured quantities do better (0.58 and 0.43), and we deliberately do not rank them against each other: at nine cells they swap places when seeds are added, which is itself the evidence that nine cells do not resolve them.

We read the result as forgetting scaling with what the new task demands of the model, and state it as the best available account rather than a demonstrated mechanism, because DMControl does not follow it — task-B difficulty there rises from minimum to maximum while forgetting falls after the medium level.

d_{trans} is not a drop-in replacement for the axis, and its leading ρ should not be read as one. Between families it separates cleanly, and that separation is what most of the correlation is made of; *within* a family it mostly does not separate at all. Gymnasium's three levels span about seven units in the median while individual cells span fifteen to twenty-three across seeds, and at twice the budget its medium and maximum medians swap places. What survives is narrower and still worth stating: a measured distance is answerable to evidence in a way a label is not, and this one answered — including about itself.

Family	Level	RD	d_{trans}	Task-B difficulty
MiniGrid	min	33.73	22.35	2.01
	med	60.88	20.64	42.60
	max	32.11	25.19	23.59
Gymnasium	min	77.45	28.75	24.63
	med	85.91	30.86	27.28
	max	64.01	24.08	17.44
DMControl	min	1.15	1.87	15.43
	med	71.64	7.58	34.07
	max	52.61	7.46	40.54
Spearman ρ with RD (predictor)		0.00 *labelled level*	0.58 *measured*	0.43 *measured*

Table 2: The nine cells behind the rank correlations, and the correlations. Task-B difficulty is the held-out reconstruction error a model reaches on task B after training on it.

Family	Level	Fine-tuning	Replay	EWC	Prog. Nets	UG-MTM
MiniGrid	min	811	1.31	800	797	1.00
	med	829	1.18	848	825	1.00
	max	32	1.02	31	31	1.00
Gymnasium	min†	0.94	0.90	0.94	0.97	1.00
	med†	1.04	0.92	1.05	1.06	1.00
	max	14	1.04	15	14	1.00
DMControl	min†	1.02	1.00	1.02	1.02	1.00
	med	16	0.37	16	16	1.00
	max	17	0.51	17	16	1.00

Table 3: Held-out task-A reconstruction error after training on task B, as a factor of the same model's error before it. 1.00 means nothing was lost. † marks control cells.

3.4 Three cells produce no forgetting, and we report them as controls

We call a cell a *control* when no method's held-out task-A reconstruction grows by more than 10% after training on task B. The threshold is not a knife edge: in every cell that does produce forgetting, the worst method ends between 14.8× and 848× its original error. Three of the nine qualify: Gymnasium at minimum and medium distance, where the task pair is the same HalfCheetah under different gravities, and DMControl at minimum. That is what the bottom of a distance axis should look like, and the benchmark detects it; it also means those cells discriminate on nothing, and the effective grid is six cells rather than nine. The criterion is deliberately blind to RD, and Gymnasium's medium cell shows why: it loses nothing in pixels and simultaneously carries the highest RD in its family.

3.5 Forgetting happens in the encoder, where the metrics cannot see it

Table 3 shows the size of the blind spot that Section 2 declared. Fine-tuning's held-out task-A reconstruction grows by a factor of 811 in MiniGrid at minimum distance while its PF for that cell is -5.58: measured in the frozen latent basis, the model *improved*. This is not an isolated sign error — fine-tuning's PF is negative in seven of the nine cells while its encoder is destroyed in six.

EWC makes the dissociation sharpest, and it was derived before it was measured. Its Fisher information is defined over $\log P(z' \mid z, a)$, in which no encoder parameter appears, so its entries there are *exactly* zero and the quadratic penalty cannot constrain them even in principle. Both halves hold: EWC preserves the transition component almost perfectly (-0.03) while its encoder degrades by a factor of 800, marginally worse than the unregularised baseline. Replay is the only method that keeps both, its task-A reconstruction never growing by more than 31%.

The lesson generalises past our metrics: **any latent-space forgetting metric evaluated on frozen representations measures the component it was pointed at and reports nothing about the representation**, and in an end-to-end world model the representation is where most of the damage is. Recording a representation-space quantity alongside costs one forward pass per evaluation.

3.6 What the benchmark says about the methods

Read across both scales the three strategies separate, and none dominates. **Replay** keeps both components and has the lower RD in five of the six cells that produce forgetting, four of them unanimously across ten seeds ($p = 0.002$, d_z from -2.28 to -3.11); it is worse in one (Gymnasium at maximum, $p = 0.006$, $d_z = +1.14$) and indistinguishable in another ($p = 0.68$). It pays where it works hardest, with the worst forward transfer of the four RSSM methods in the cells where it best protects the encoder — though replay trains on A and B together, so half its gradient steps go to task-A data and its FT confounds transfer with a halved budget on B. **Regularisation** protects what its penalty covers and nothing else, as above. **Structural isolation**, ours included, buys preservation by declining to learn: our method's task-A reconstruction is unchanged to the bit in all nine cells, because its encoder is frozen, and it pays in forward transfer by two orders of magnitude more than any other method. Freezing has a second consequence we did not anticipate: the post-B transition models become overconfident, with some latent dimensions collapsing to $\sigma = 0.007$ and the rollout KL already at 80 at step 0. Confident and wrong is a failure mode an unbounded divergence surfaces and a mean hides.

4 Discussion

4.1 What "controlled dynamic distance" can and cannot mean

What is genuinely controlled is the *construction*, and in Gymnasium in the strongest sense available: the maximum level is the medium perturbation plus two further ones on the same task pair. If the axis ordered anything monotone, a strict superset could not come out below. What is not controlled is the quantity the label is taken to stand for, and the ordering of the labels survives contact with neither of the two things we can measure. A label that neither predicts the outcome nor tracks the measured construct is a name, not a control.

The recommendation is narrow and cheap: **report a measured distance for every task pair and treat level labels as construction parameters rather than as an independent variable.** d_{trans} costs one model per environment — a 20% surcharge on our training budget —

and is defined wherever transitions can be sampled. It is not a finished instrument, as Section 3.3 reports of our own, but it is measured, comparable, and answerable.

4.2 Two things called "distance" are being conflated

The peak at the medium level repeats across families, which invites a mechanism rather than a list of exceptions. The one we favour, and cannot yet establish, is that forgetting tracks *how much the model actually updates*, and that this is non-monotone in distance for a reason unrelated to similarity: at minimum distance there is little to update, and at maximum distance the model often fails to fit task B at all. Overwriting needs a task that is both different and learnable.

If that is right, task distance bundles two variables that come apart. *Dissimilarity* is how far the new dynamics sit from the old. *Learning demand* is how much the model must change to fit the new task at the available budget. They correlate at the low end and decouple at the high end, where a distant task can be too hard to learn from. That is why a single scalar predictor reaches 0.58 and not higher, and why DMControl breaks the account: there difficulty has stopped meaning "much to learn" and started meaning "too hard to learn". Separating them needs a predictor built from what the optimiser did during task B rather than from the outcome, which we did not run.

4.3 What the encoder result asks of benchmark design

EWC's zero is worth stating generally: **the protected set is exactly the support of whatever likelihood the Fisher is estimated from**, and in modular architectures that support is usually narrower than the module list suggests. The failure is invisible to any metric sharing the same restriction — which, in our first formulation, was every metric we reported. A component-level benchmark that reports only its component-level metric can certify the preservation of a system that has been substantially destroyed.

4.4 Limitations

The binding constraint is statistical resolution: at ten seeds the exact paired test floors at $p \approx 0.002$, and the three control cells remain at five, where the floor is 0.0625. The grid is nine cells of which six discriminate, and one architecture family — every method here is a variation on the same recurrent latent world model, so the results speak to that class. The experiments reported here run *one* task switch, which is a thin reading of "continual"; a four-task MiniGrid sequence, reported in full elsewhere, shows the same intermediate peak along sequence position rather than a monotone increase, but that is one family and one sequence. The budget (20 random-policy episodes per task, 5000 updates, latent dimension 32) is small enough that absolute magnitudes are budget-dependent, though task A is demonstrably learned at this scale and the central finding survives doubling it. Task-B difficulty and RD are computed from the same runs, so a cell with hard data inflates both; this is not circular but admits the reading "this cell is hard". Finally, the prediction that distinguishes our account from the design's is cheap and untested: forgetting should fall again at distances large enough that task B stops being learnable at the given budget.

5 Conclusion

We built a benchmark to ask which continual learning method best preserves a world model's transition component, and the two things it told us are about the question rather than the answer. The distance axis it was designed around does not order forgetting in any of three families, and the label carries no rank information across the nine cells; what forgetting tracks is closer to what the new task demands than to how far it sits from the old. And the forgetting is concentrated in the encoder, where component-level metrics — ours included — are blind by construction, so a suite reporting only them would certify the preservation of models whose representation had been destroyed eight-hundred-fold. Both results are cheap to act on: report a measured distance per task pair and hold it to account, and report a representation-space quantity beside the component-level one. We characterise five methods rather than ranking them, our own among them and not favourably.

Reproducibility

Code, configurations, the table-generating scripts and all 375 result files are released. Every table here is generated from the stored results rather than transcribed; the protocol is declared in configuration, recorded in every result file, and the runner refuses to combine results produced under different ones. A cell re-executed eight days after its original run reproduced to the fourth decimal.

References

[1] David Ha and Jürgen Schmidhuber. World models. *arXiv preprint arXiv:1803.10122*, 2018.

[2] Danijar Hafner, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. Learning latent dynamics for planning from pixels. In *International Conference on Machine Learning (ICML)*, 2019.

[3] Samuel Kessler, Mateusz Ostaszewski, Michał P. Bortkiewicz, Mateusz Żarski, Maciej Wołczyk, Jack Parker-Holder, Stephen J. Roberts, and Piotr Miłoś. The effectiveness of world models for continual reinforcement learning. In *Conference on Lifelong Learning Agents (CoLLAs)*, volume 232 of *PMLR*, pages 184–204, 2023.

[4] James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, Demis Hassabis, Claudia Clopath, Dharshan Kumaran, and Raia Hadsell. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13):3521–3526, 2017.

[5] Andrei A. Rusu, Neil C. Rabinowitz, Guillaume Desjardins, Hubert Soyer, James Kirkpatrick, Koray Kavukcuoglu, Razvan Pascanu, and Raia Hadsell. Progressive neural networks. *arXiv preprint arXiv:1606.04671*, 2016.

[6] Gido M. van de Ven, Nicholas Soures, and Dhireesha Kudithipudi. Continual learning and catastrophic forgetting. *arXiv preprint arXiv:2403.05175*, 2024.

[7] Maciej Wołczyk, Michał Zając, Razvan Pascanu, Łukasz Kuciński, and Piotr Miłoś. Continual world: A robotic benchmark for continual reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.